

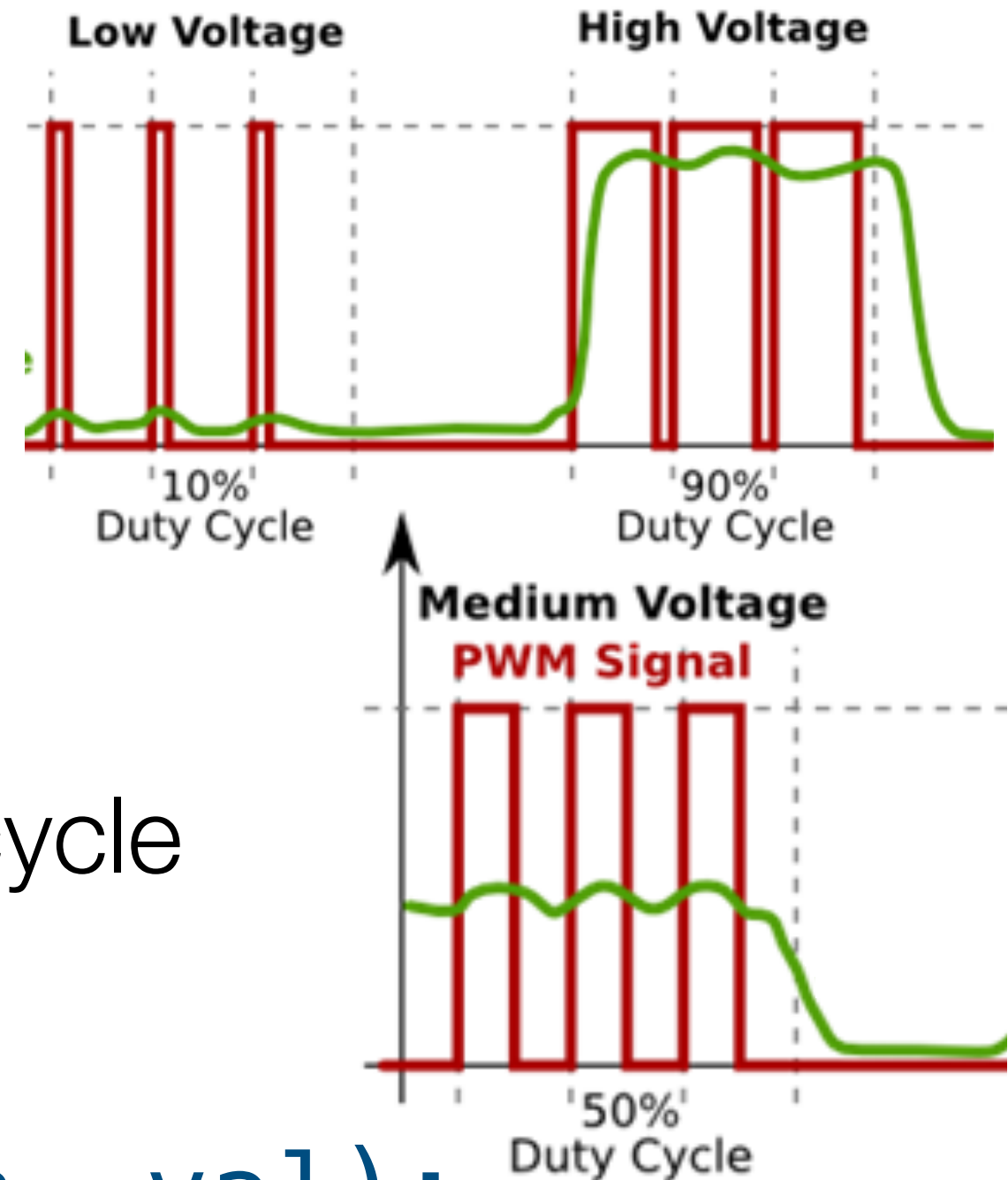
EG1311: Design & Make

PWM & Servo

Dr. Jason S. Ku

Analog Output

- Want voltage 0V - 5V?
- Turn on and off fast!
- Pulse Width Modulation
- Average voltage = duty cycle



```
void analogWrite(pin, val);
```

- PWM pins 3,5,6,9,10,11 (~)
- Write 8-bit value **[0, 255]**

Analog Demo

```
int motor_pin = 3;    // (~) PWM pin!

void setup() {
}

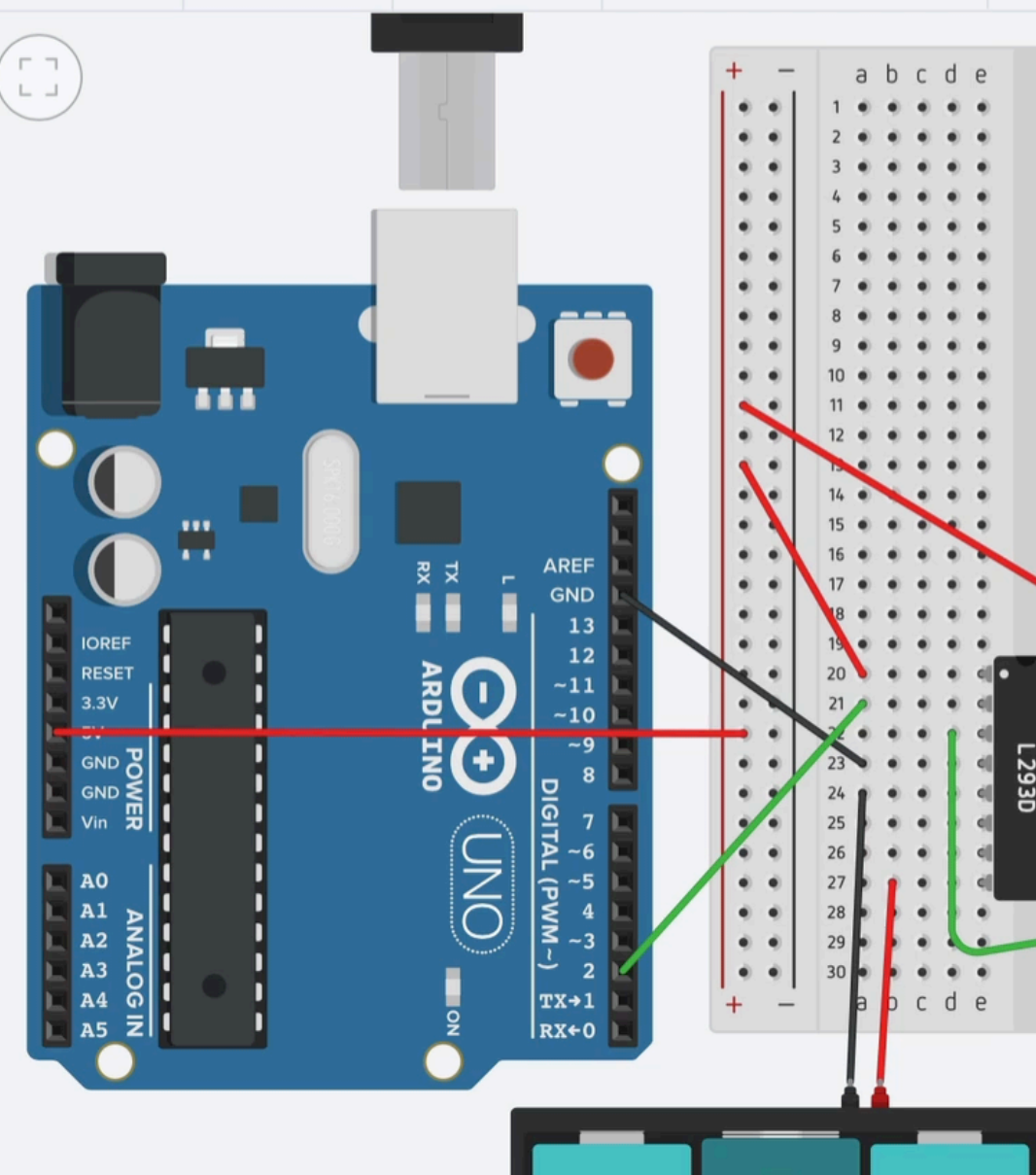
void loop() {
    for (int i = 0; i < 256; i++) {
        analogWrite(motor_pin, i);
        delay(10);
    }
    for (int i = 255; i >= 0; i--) {
        analogWrite(motor_pin, i);
        delay(10);
    }
}
```

Analog Demo

TIN KER CAD Daring Bombul-Jarv All changes saved

Code Start Simulation Export Share

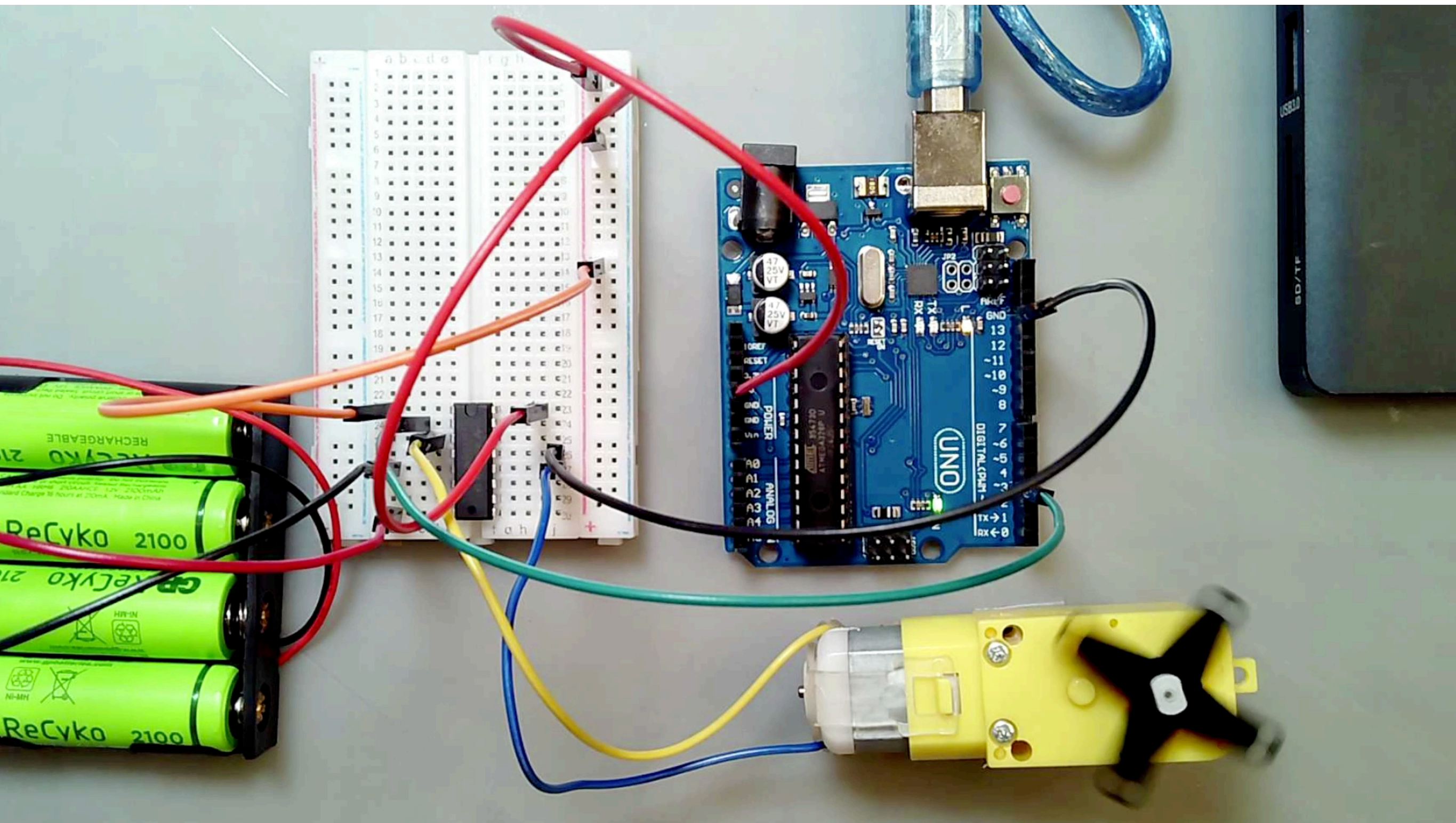
Text 1 (Arduino Uno R3)



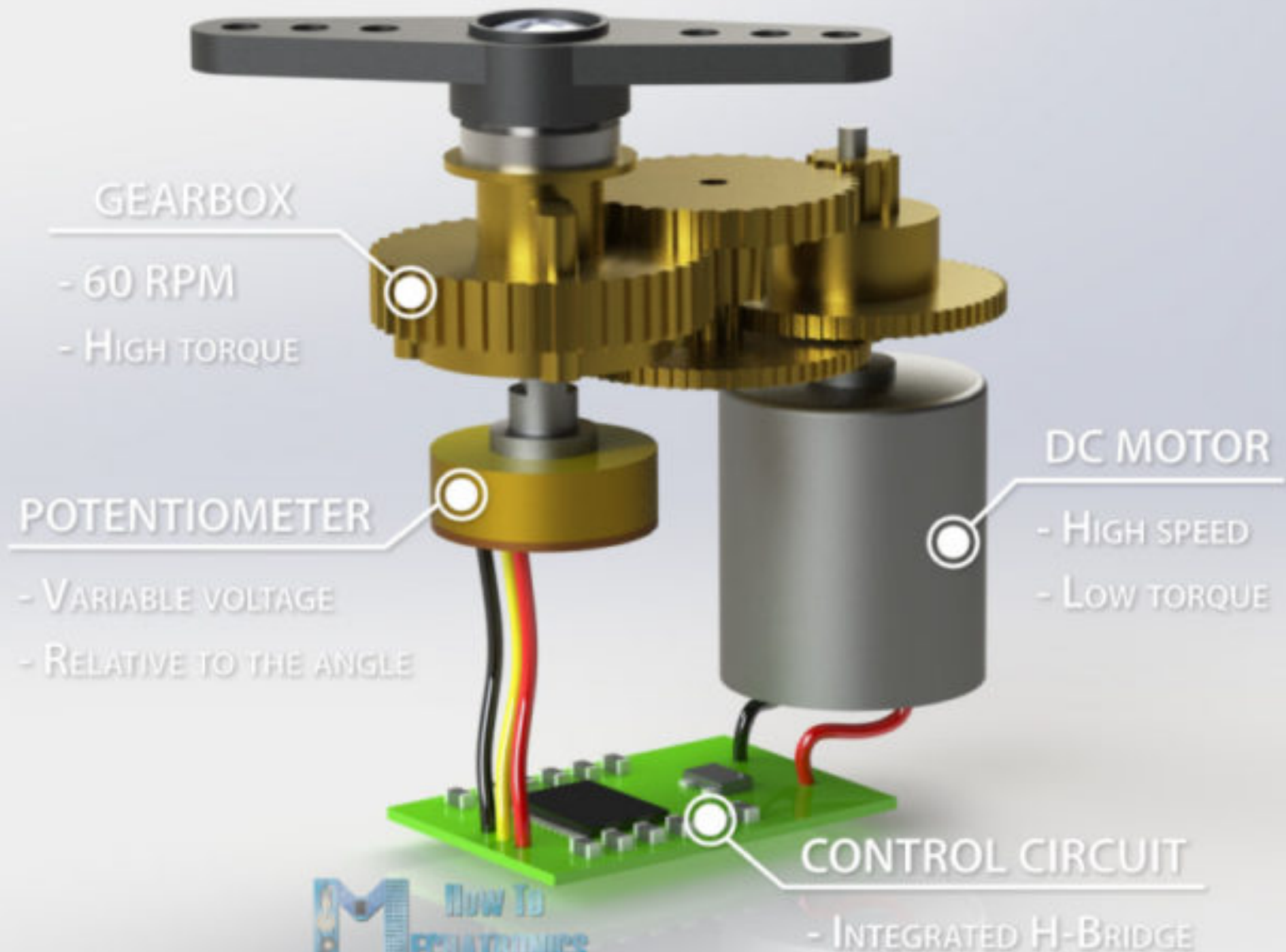
```
1 int motor_pin = 3;  // (~) PWM pin!
2
3 void setup() {
4 }
5
6 void loop() {
7   for (int i = 0; i < 256; i++) {
8     analogWrite(motor_pin, i);
9     delay(10);
10  }
11  for (int i = 255; i >= 0; i--) {
12    analogWrite(motor_pin, i);
13    delay(10);
14  }
15 }
```

Serial Monitor

Analog Demo

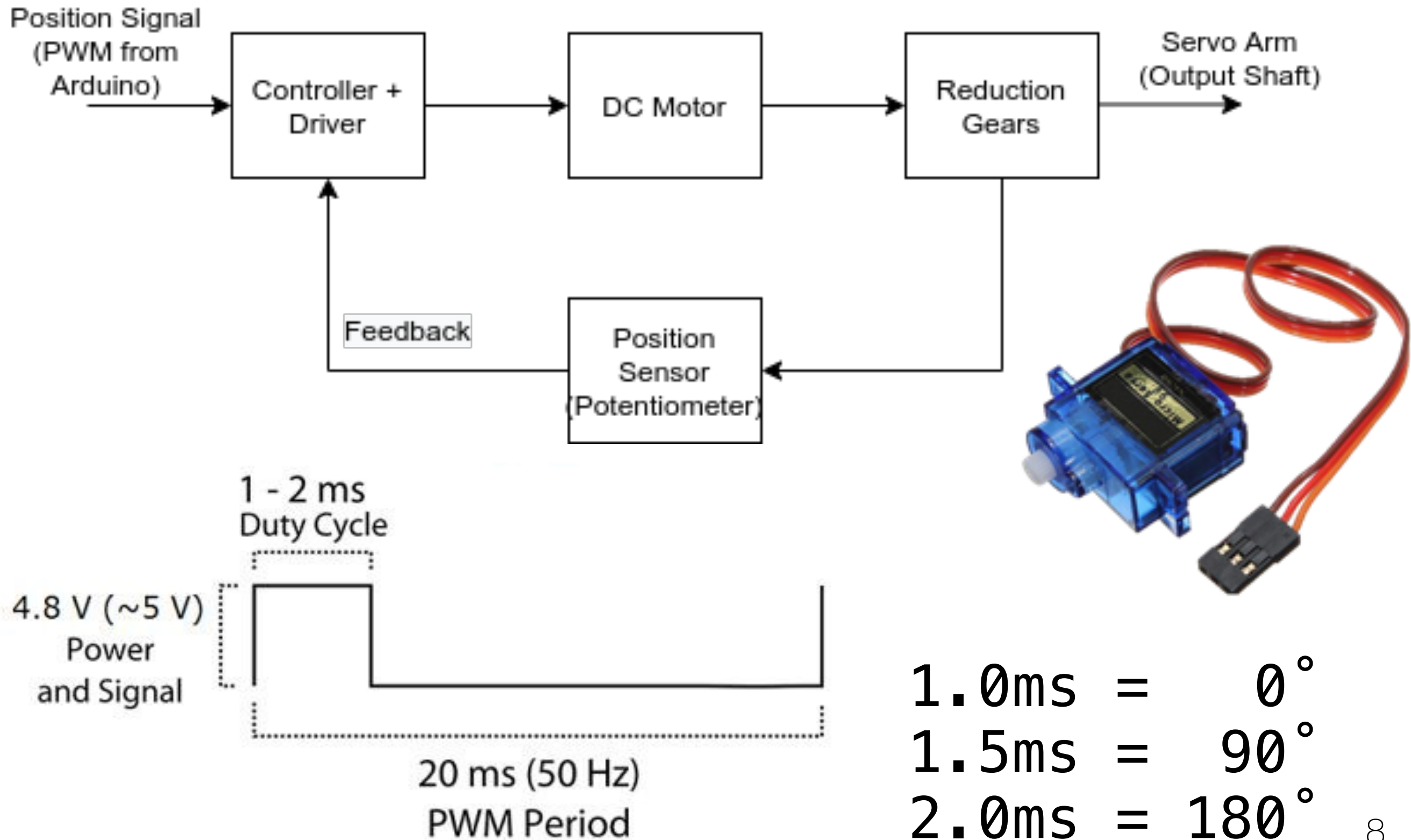
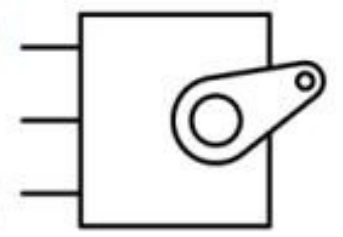


Servo Motor



Servo Motor

PWM=Orange (⏏)
Vcc=Red (+)
Ground=Brown (-)



Servo Motor Library

- Takes away PWM timer from pins 9,10
- Do not use PWM on 9,10 if using servo!

```
#include <Servo.h>
```

```
Servo servo;
```

```
servo.attach(pin, 1000, 2000);
```

```
servo.write(deg);
```

1.0ms = 0°

1.5ms = 90°

2.0ms = 180°

Servo Motor Library

- Takes away PWM timer from pins 9,10
- Do not use PWM on 9,10 if using servo!

```
#include <Servo.h>
```

```
Servo servo;
```

```
servo.attach(pin, 660, 2400);
```

```
servo.write(deg);
```

~~1.0ms = 0°
1.5ms = 90°
2.0ms = 180°~~

Servo Demo

```
#include <Servo.h>
Servo servo;
int servo_pin = 2;  // can be any digital pin!

void setup() {
    servo.attach(servo_pin, 660, 2400);
}

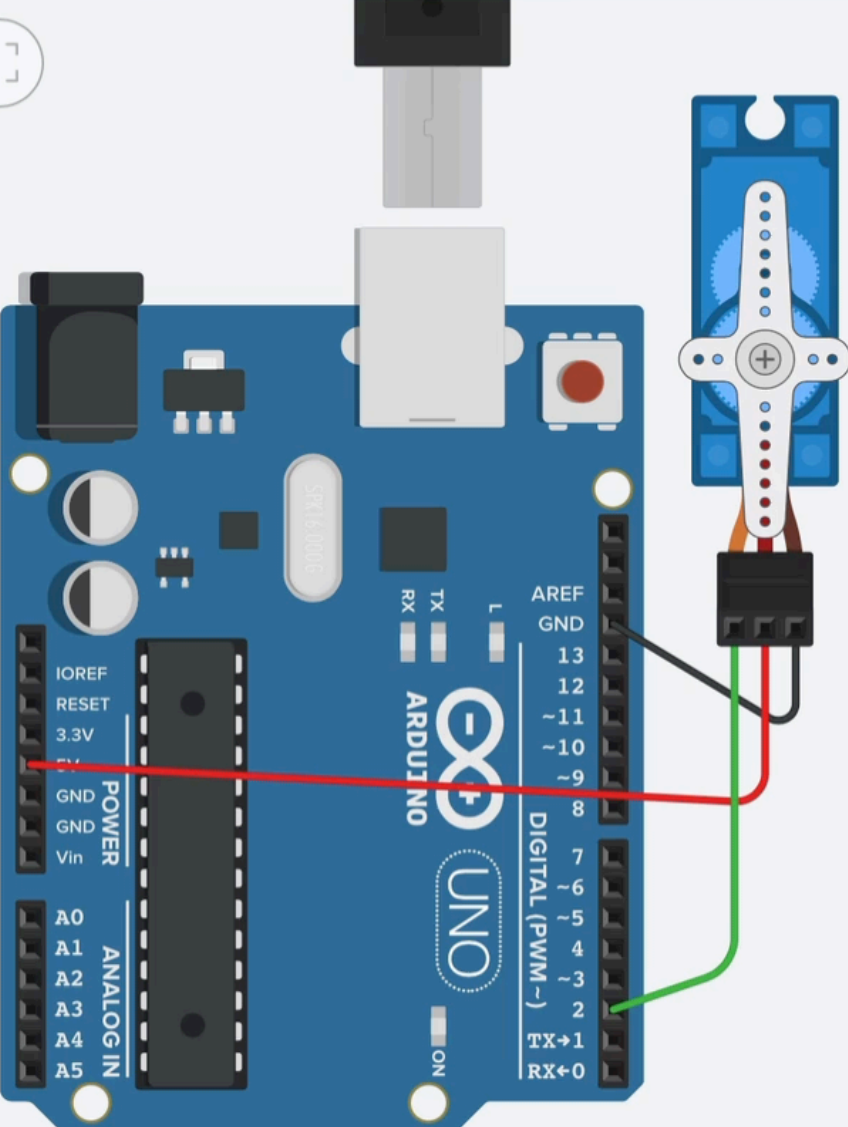
void loop() {
    for (int i = 0; i <= 180; i++) {
        servo.write(i);
        delay(5);
    }
    for (int i = 180; i >= 0; i--) {
        servo.write(i);
        delay(5);
    }
}
```

Servo Demo

TINKER CAD Daring Bombul-Jarv All changes saved

Code Start Simulation Export Share

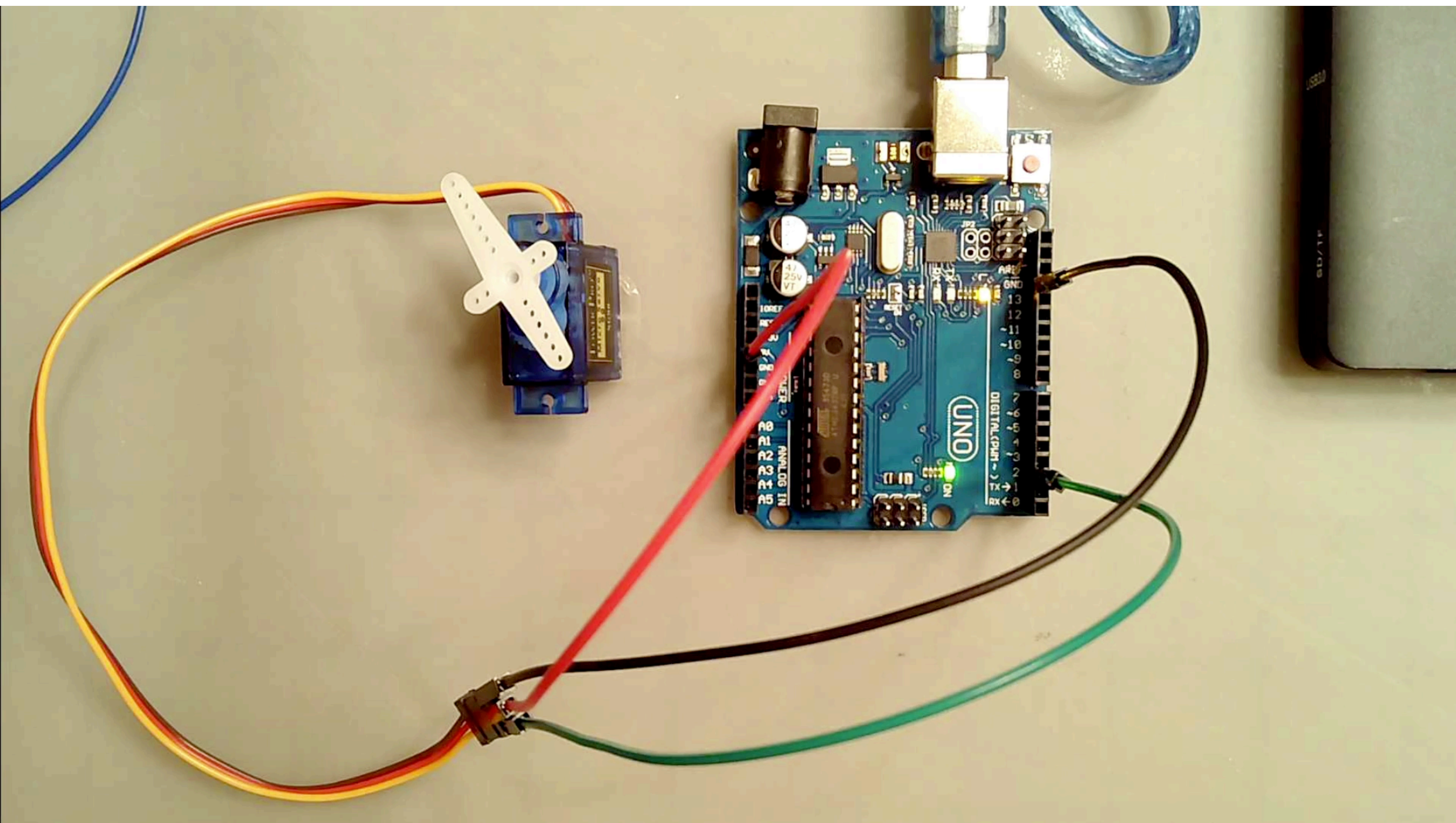
Text 1 (Arduino Uno R3)



```
1 #include <Servo.h>
2 Servo servo;
3 int servo_pin = 2; // can be any digital pin!
4
5 void setup() {
6     servo.attach(servo_pin, 300, 2700);
7 }
8
9 void loop() {
10     for (int i = 0; i <= 180; i++) {
11         servo.write(i);
12         delay(10);
13     }
14     for (int i = 180; i >= 0; i--) {
15         servo.write(i);
16         delay(10);
17     }
18 }
```

Serial Monitor

Servo Demo



Practice

After calling the function `analogWrite(3, 100);` on an Arduino UNO, the average voltage on the PWM pin 3 (relative to ground) will be closest to:

Respond at pollev.com/jku

A

B

C

D

A) 1 Volts

B) 2 Volts

C) 3 Volts

D) 4 Volts

Practice

Which Arduino UNO PWM pins are disabled when using the Servo library?

Respond at pollev.com/jku

A

B

C

D

A) Pins 0 & 1

B) Pins 5 & 6

C) Pins 9 & 10

D) Pins 3 & 11

EG1311: Design & Make

PWM & Servo

Dr. Jason S. Ku